Ex. threshold of 2.5V, 5V supply, 10kΩ resistors

$V = IR$ $\quad R_{Total} = R_1 + R_2$ $\quad V_{thresh} = IR_2$ (across R_2)

$I = \frac{V}{R} = \frac{V_{supply}}{R_1 + R_2}$ and $V_{Thresh} = V_{supply}\left(\frac{R_2}{R_1 + R_2}\right)$

$V_{thresh} = 2.5 = 5\left(\frac{R_2}{R_1 + R_2}\right) \longrightarrow R_1 = R_2$

$R_{Total} = 10k\Omega + 10k\Omega = 20k\Omega$

$I = \frac{5V}{20000\Omega} = 0.00025A = 0.25mA$

$V_{R_2} = IR_2 = (0.00025)(10{,}000) = 2.5V = V_{Thresh}$

$V_{R_1} = IR_1 = (0.00025)(10000) = 2.5V$

$2.5V + 2.5V = 5V_{supply}$

Testing comparator w/ values of 2V and 3V inputs:

$V_+ = V_{in}$
$V_- = V_{Thresh} = 2.5V$

2V: $V_{diff} = V_+ - V_- = V_{in} - 2.5 = 2 - 2.5 = -0.5V$ so LOW (under threshold)

3V: $V_{diff} = V_+ - V_- = V_{in} - 2.5 = 3 - 2.5 = 0.5V$ so HIGH (over threshold)